knocking out all known sources of measurement bias, and still finding numbers that ever more precisely do not agree with each other.

It's unclear what the solution to this problem will end up being. Maybe it really does come down to systematic errors in the data, or some problem with the measurements themselves. Maybe it's just a statistical fluke, as unlikely as that looks on the surface. Some of the most intriguing explanations involve dark energy that is not your garden-variety cosmological constant, but is instead something rather more ominous—something that could perhaps lead to a Big Rip. There's one hypothesis that would go a reasonable way toward fixing the discrepancy between the measurements: dark energy getting more powerful over time, in just the way you might expect from the early stages of a phantom-dark-energy-dominated cosmos.

We probably shouldn't panic just yet. As discussed, the data still aren't that clear. Most measurements of w give a value that is fully consistent with -1, and though it's true that values less than -1 are sometimes very slightly preferred, that preference isn't really statistically meaningful. As for the Hubble Constant disagreement, even if all the measurements are correct, nonapocalyptic explanations for the discrepancy—involving weird models of dark matter, or altered conditions in the early universe—are very much in the running. In fact, even tweaking dark energy wouldn't be enough to totally solve the problem, so it's not unreasonable to assume that the solution might lie elsewhere. And even if there *has* been a sharp upturn in the effects of dark energy in recent cosmic history, suggesting something like phantom dark energy, we still have a LOT of time before a Big Rip could possibly occur.

In fact, the one thing that all the universe-ending scenarios we've already discussed have in common is that they definitely aren't coming around anytime soon. As far as we can tell from our best understanding of physics, we have at least tens of billions of years before even the most extreme version of a sudden Big Crunch reversal could occur, and no Big Rip could be less than a hundred billion years off. A Heat Death, considered by most to be even more likely, would be so far into the cosmic depths of the future that we hardly have terms to describe it.

There is one possibility, though, that is decidedly more menacing than all the rest. It presents the prospect of a doomsday brought upon us by, in essence, a manufacturer's defect in the fabric of the cosmos itself. It's